Quiz: Indexes (1)

Name:

Quiz Rules:

1. you MAY use an printed or hand written material
2. you MAY NOT use a computer or other electronic device

All of the problems assume the following schema.

```
CREATE TABLE users (
    id_users BIGINT UNIQUE NOT NULL,
    created_at TIMESTAMPTZ,
    username TEXT NOT NULL
);

CREATE TABLE tweets (
    id_tweets BIGINT PRIMARY KEY,
    id_users BIGINT REFERENCES users(id_users),
    in_reply_to_user_id BIGINT REFERENCES users(id_users),
    created_at TIMESTAMPTZ,
    text TEXT
);

CREATE TABLE tweet_tags (
    id_tweets BIGINT REFERENCES tweets(id_tweets),
    tag TEXT,
    UNIQUE(id_tweets, tag)
);
```

1. Recall that certain constraints create indexes on the appropriate columns. List the equivalent `CREATE INDEX` commands that are run by these constraints.

   Note: When solving all subsequent problems, assume only the indexes above have been created. For example, if problem 3 asks you to create an index, you should not assume that index has been created when working on problem 4.

2. List the scan methods applicable for the following SQL query.

```
SELECT count(*)
FROM tweets
WHERE in_reply_to_user_id IS NOT NULL
  AND id_users = :input_user_id
```

3. Create index(es) so that the following query can use an index only scan.

   Do not create any unneeded indexes; if no new indexes are needed, say so.

```
SELECT count(*)
FROM tweet_tags
WHERE lower(tag)=:tag;
```

4. Create index(es) so that the following query can use an index only scan, avoid an explicit sort, and take advantage of the LIMIT clause for faster processing.

   Do not create any unneeded indexes; if no new indexes are needed, say so.

```
SELECT tag
FROM tweet_tags
WHERE id_tweets = :id_tweets
  AND tag >= :start_tag
  AND tag <= :end_tag
ORDER BY tag ASC
LIMIT 2;
```

5. Create index(es) so that the following query will run as efficiently as possible.

   Do not create any unneeded indexes; if no new indexes are needed, say so.

```
SELECT created_at
FROM tweet_tags
JOIN tweets USING (id_tweets)
WHERE tag = :tag
  AND id_users = :id_users
ORDER BY id_tweets DESC
LIMIT 10;
```